14

General Linear Model

The general linear model incorporates many of the most popular and useful models that arise in applied statistics, including models for multiple regression and the analysis of variance. The basic model can be written succinctly in matrix form as

$$Y = X\beta + \epsilon, \tag{14.1}$$

where Y, our observed data, is a random vector in $\mathbb{R}^n$, X is an $n \times p$ matrix of known constants, $\beta \in \mathbb{R}^p$ is an unknown parameter, and ϵ is a random vector in $\mathbb{R}^n$ of unobserved errors. We usually assume that $\epsilon_1, \dots, \epsilon_n$ are a random sample from $N(0, \sigma^2)$, with $\sigma > 0$ an unknown parameter, so that

$$\epsilon \sim N(0, \sigma^2 I). \tag{14.2}$$

But some of our results hold under the less restrictive conditions that $E\epsilon_i = 0$ for all i, $\mathrm{Var}(\epsilon_i) = \sigma^2$ for all i, and $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$. In matrix notation, $E\epsilon = 0$ and $\mathrm{Cov}(\epsilon) = \sigma^2 I$. Since Y is ϵ plus a constant vector and $E\epsilon = 0$, we have $EY = X\beta$ and $\mathrm{Cov}(Y) = \mathrm{Cov}(\epsilon) = \sigma^2 I$. With the normal distribution for ϵ in (14.2),

$$Y \sim N(X\beta, \sigma^2 I). \tag{14.3}$$

Example 14.1 (Quadratic Regression). In quadratic regression, a response variable Y is modeled as a quadratic function of some explanatory variable x plus a random error. Specifically,

$$Y_i = \beta_1 + \beta_2 x_i + \beta_3 x_i^2 + \epsilon_i, \qquad i = 1, \dots, n.$$

Here the explanatory variables $x_1, \dots, x_n$ are taken to be known constants, β_1, β_2, and β_3 are unknown parameters, and $\epsilon_1, \dots, \epsilon_n$ are i.i.d. from $N(0, \sigma^2)$. If we define the design matrix X as

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$$X = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{pmatrix},$$

then $Y = X\beta + \epsilon$, as in (14.1).

Example 14.2 (One-Way ANOVA). Suppose we have independent random samples from three normal populations with common variance σ^2, so

$$Y_i \sim \begin{cases} N(\beta_1, \sigma^2), & i = 1, \dots, n_1; \\ N(\beta_2, \sigma^2), & i = n_1 + 1, \dots, n_1 + n_2; \\ N(\beta_3, \sigma^2), & i = n_1 + n_2 + 1, \dots, n_1 + n_2 + n_3 \stackrel{\text{def}}{=} n. \end{cases}$$

If we define

$$X = \begin{pmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \end{pmatrix},$$

then $EY = X\beta$ and the model has form (14.3).

In applications the parameters $\beta_1, \dots, \beta_p$ usually arise naturally when formulating the model. As a consequence they are generally easy to interpret. But for technical reasons it is often more convenient to view the unknown mean of Y, namely,

$$\xi \stackrel{\text{def}}{=} EY = X\beta$$

in $\mathbb{R}^n$ as the unknown parameter. If $c_1, \dots, c_p$ are the columns of X, then

$$\xi = X\beta = \beta_1 c_1 + \cdots + \beta_p c_p,$$

which shows that ξ must be a linear combination of the columns of X. So ξ must lie in the vector space

$$\omega \stackrel{\text{def}}{=} \operatorname{span}\{c_1, \dots, c_p\} = \{X\beta : \beta \in \mathbb{R}^p\}.$$

Using ξ instead of β, the vector of unknown parameters is $\theta = (\xi, \sigma)$ taking values in $\Omega = \omega \times (0, \infty)$.

Since Y has mean ξ, it is fairly intuitive that our data must provide some information distinguishing between any two values for ξ, since the distributions for Y under two different values for ξ must be different. Whether this also holds for β depends on the rank r of X. Since X has p columns, this rank r is at most p. If the rank of X equals p then the mapping $\beta \rightsquigarrow X\beta$ is one-to-one, and each value $\xi \in \omega$ is the image of a unique value $\beta \in \mathbb{R}^p$. But if the columns of X are linearly dependent, then a nontrivial linear combination of the columns of X will equal zero, so $Xv = 0$ for some $v \neq 0$. But then $X(\beta + v) = X\beta + Xv = X\beta$, and parameters β and $\beta^* = \beta + v$ both give the same mean ξ. Here our data Y provides no information to distinguish between parameter values β and β^*.

Example 14.3. Suppose

$$X = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

Here the three columns of X are linearly dependent because the first column is the sum of the second and third columns, and the rank of X is 2, $r = 2 < p = 3$. Note that parameter values

$$\beta = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \beta^* = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

both give

$$\xi = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

14.1 Canonical Form

Many results about testing and estimation in the general linear model follow easily once the data are expressed in a canonical form. Let $v_1, \dots, v_n$ be an orthonormal basis for $\mathbb{R}^n$, chosen so that $v_1, \dots, v_r$ span ω. Entries in the canonical data vector Z are coefficients expressing Y as a linear combination of these basis vectors,

$$Y = Z_1 v_1 + \cdots + Z_n v_n. \tag{14.4}$$

Algebraically, Z can be found introducing an $n \times n$ matrix O with columns $v_1, \dots, v_n$. Then O is an orthogonal matrix, $O'O = OO' = I$, and Y and Z are related by

$$Z = O'Y \text{ or } Y = OZ.$$

Since $Y = \xi + \epsilon$, $Z = O'(\xi + \epsilon) = O'\xi + O'\epsilon$. If we define $\eta = O'\xi$ and $\epsilon^* = O'\epsilon$, then

$$Z = \eta + \epsilon^*.$$

Because $E\epsilon^* = EO'\epsilon = O'E\epsilon = 0$ and

$$\mathrm{Cov}(\epsilon^*) = \mathrm{Cov}(O'\epsilon) = O'\mathrm{Cov}(\epsilon)O = O'(\sigma^2 I)O = \sigma^2 O'O = \sigma^2 I,$$

$\epsilon^* \sim N(0, \sigma^2 I)$ and $\epsilon_1^*, \dots, \epsilon_n^*$ are i.i.d. from $N(0, \sigma^2)$. Since $Z = \eta + \epsilon^*$,

$$Z \sim N(\eta, \sigma^2 I). \tag{14.5}$$

Next, let $c_1, \dots, c_p$ denote the columns of the design matrix X. Then $\xi = X\beta = \sum_{i=1}^p \beta_i c_i$ and

$$\eta = O'\xi = \begin{pmatrix} v_1' \\ \vdots \\ v_n' \end{pmatrix} \sum_{i=1}^p \beta_i c_i = \begin{pmatrix} \sum_{i=1}^p \beta_i v_1' c_i \\ \vdots \\ \sum_{i=1}^p \beta_i v_n' c_i \end{pmatrix}.$$

Since $c_1, \dots, c_p$ all lie in ω, and $v_{r+1}, \dots, v_n$ all lie in $\omega^\perp$, we have $v_k' c_i = 0$ for $k > r$, and thus

$$\eta_{r+1} = \cdots = \eta_n = 0. \tag{14.6}$$

Because $\eta = O'\xi$,

$$\xi = O\eta = (v_1 \dots v_n) \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_r \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \sum_{i=1}^r \eta_i v_i.$$

These equations establish a one-to-one relation between points $\xi \in \omega$ and $(\eta_1, \dots, \eta_r) \in \mathbb{R}^r$.

Since $Z \sim N(\eta, \sigma^2 I)$, the variables $Z_1, \dots, Z_n$ are independent with $Z_i \sim N(\eta_i, \sigma^2)$. The density of Z, taking advantage of the fact that $\eta_{r+1} = \cdots = \eta_n = 0$, is

$$\begin{aligned} &\frac{1}{\sqrt{2\pi\sigma^2}^n} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^r (z_i - \eta_i)^2 - \frac{1}{2\sigma^2} \sum_{i=r+1}^n z_i^2\right] \\ &\qquad = \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n z_i^2 + \frac{1}{\sigma^2} \sum_{i=1}^r \eta_i z_i - \sum_{i=1}^r \frac{\eta_i^2}{2\sigma^2} - \frac{n}{2}\log(2\pi\sigma^2)\right]. \end{aligned}$$

These densities form a full rank $(r+1)$-parameter exponential family with complete sufficient statistic

$$\left(Z_1, \dots, Z_r, \sum_{i=1}^n Z_i^2\right). \tag{14.7}$$

14.2 Estimation

Exploiting the canonical form, many parameters are easy to estimate. Because $EZ_i = \eta_i$, $i = 1, \dots, r$, Z_i is the UMVU estimator of η_i, $i = 1, \dots, r$. Also, since $\xi = \sum_{i=1}^r \eta_i v_i$,

$$\hat{\xi} = \sum_{i=1}^{r} Z_i v_i \tag{14.8}$$

is a natural estimator of ξ. Noting that

$$E\hat{\xi} = \sum_{i=1}^{r} EZ_i v_i = \sum_{i=1}^{r} \eta_i v_i = \xi,$$

$\hat{\xi}$ is unbiased. Since it is a function of the complete sufficient statistic, it should be optimal in some sense. One measure of optimality might be the expected squared distance from the true value ξ. If $\tilde{\xi}$ is a competing unbiased estimator, then

$$E\|\tilde{\xi} - \xi\|^2 = \sum_{j=1}^{n} E(\tilde{\xi}_j - \xi_j)^2 = \sum_{j=1}^{n} \mathrm{Var}(\tilde{\xi}_j). \tag{14.9}$$

Because $\hat{\xi}_j$ is unbiased for ξ_j and is a function of the complete sufficient statistic, $\mathrm{Var}(\hat{\xi}_j) \le \mathrm{Var}(\tilde{\xi}_j)$, $j = 1, \dots, n$. So $\hat{\xi}$ minimizes each term in the sum in (14.9), and hence

$$E\|\hat{\xi} - \xi\|^2 \le E\|\tilde{\xi} - \xi\|^2.$$

A more involved argument shows that $\hat{\xi}$ also minimizes the expectation of any other nonnegative quadratic form in the estimation error, $E(\tilde{\xi} - \xi)'A(\tilde{\xi} - \xi)$, among all unbiased estimators.

From (14.4), we can write Y as

$$Y = \sum_{i=1}^{r} Z_i v_i + \sum_{i=r+1}^{n} Z_i v_i = \hat{\xi} + \sum_{i=r+1}^{n} Z_i v_i.$$

In this expression the first summand, $\hat{\xi}$, lies in ω, and the second, $Y - \hat{\xi} = \sum_{i=r+1}^{n} Z_i v_i$, lies in $\omega^\perp$. This difference $Y - \hat{\xi}$ is called the *vector of residuals*, denoted by e:

$$e \overset{\text{def}}{=} Y - \hat{\xi} = \sum_{i=r+1}^{n} Z_i v_i. \tag{14.10}$$

Since $Y = \hat{\xi} + e$, by the Pythagorean theorem, if $\tilde{\xi}$ is any point in ω, then

$$\|Y - \tilde{\xi}\|^2 = \|\hat{\xi} - \tilde{\xi} + e\|^2 = \|\hat{\xi} - \tilde{\xi}\|^2 + \|e\|^2,$$

because $\hat{\xi} - \tilde{\xi} \in \omega$ is orthogonal to $e \in \omega^\perp$. From this equation, it is apparent that $\hat{\xi}$ is the unique point in ω closest to the data vector Y. This closest point is called the *projection* of Y onto ω. The mapping $Y \rightsquigarrow \hat{\xi}$ is linear and can be represented by an $n \times n$ matrix P,

$$\hat{\xi} = PY,$$

with P called the (orthogonal) *projection matrix* onto ω. Since $\hat{\xi} \in \omega$, $P\hat{\xi} = \hat{\xi}$, and so $P^2Y = P(PY) = P\hat{\xi} = \hat{\xi} = PY$. Because Y can take arbitrary values in $\mathbb{R}^n$, this shows that $P^2 = P$. (Matrices that satisfy this equation are called *idempotent*.) Using the orthonormal basis, P can be written as $P = v_1v_1' + \cdots + v_rv_r'$. (To check that this works, just multiply (14.4) by this sum.) But for explicit calculation, formulas that do not rely on construction of the basis vectors $v_1, \dots, v_n$ are more convenient, and are developed below.

Since arbitrary points in ω can be written as $X\beta$ for some $\beta \in \mathbb{R}^p$, if $\hat{\xi} = X\hat{\beta}$, then $\hat{\beta}$ must minimize

$$\|Y - X\beta\|^2 = \sum_{i=1}^{n} \bigl[Y_i - (X\beta)_i\bigr]^2 \tag{14.11}$$

over $\beta \in \mathbb{R}^p$. For this reason, $\hat{\beta}$ is called the *least squares estimator* of β. Of course, when the rank r of X is less than p, $\hat{\beta}$ will not be unique. But unique or not, all partial derivatives of the least squares criterion (14.11) must vanish at $\beta = \hat{\beta}$. This often provides a convenient way to calculate $\hat{\beta}$ and then $\hat{\xi}$.

Another approach to explicit calculation proceeds directly from geometric considerations. Since the columns c_i, $i = 1, \dots, p$, of X lie in ω, and $e = Y - \hat{\xi}$ lies in $\omega^\perp$, we must have $c_i'e = 0$, which implies

$$X'e = 0.$$

Since $Y = \hat{\xi} + e$,

$$X'Y = X'(\hat{\xi} + e) = X'\hat{\xi} + X'e = X'\hat{\xi} = X'X\hat{\beta}. \tag{14.12}$$

If $X'X$ is invertible, then this equation gives

$$\hat{\beta} = (X'X)^{-1}X'Y. \tag{14.13}$$

The matrix $X'X$ is invertible if X has full rank, $r = p$. In fact, $X'X$ is positive definite in this case. To see this, let v be an eigenvector of $X'X$ with $\|v\| = 1$ and eigenvalue λ. Then

$$\|Xv\|^2 = v'X'Xv = \lambda v'v = \lambda,$$

which must be strictly positive since $Xv = c_1v_1 + \cdots + c_pv_p$ cannot be zero if X has full rank. When X has full rank

$$PY = \hat{\xi} = X\hat{\beta} = X(X'X)^{-1}X'Y,$$

and so the projection matrix P onto ω can be written as

$$P = X(X'X)^{-1}X'. \tag{14.14}$$

Since $\hat{\xi}$ is unbiased, $a'\hat{\xi}$ is an unbiased estimator of $a'\xi$. This estimator is UMVU because $\hat{\xi}$ is a function of the complete sufficient statistic. By (14.12), $X'Y = X'\hat{\xi}$, and so by (14.13), when X is full rank

$$\hat{\beta} = (X'X)^{-1}X'\hat{\xi}.$$

This equation shows that $\hat{\beta}_i$ is a linear function of $\hat{\xi}$, and so $\hat{\beta}_i$ is UMVU for β_i.

14.3 Gauss–Markov Theorem

For this section we relax the assumptions for the general linear model. The model still has $Y = X\beta + \epsilon$, but now the ϵ_i, $i = 1, \dots, n$, need not be a random sample from $N(0, \sigma^2)$. Instead, we assume the ϵ_i, $i = 1, \dots, n$, have zero mean, $E\epsilon_i = 0$, $i = 1, \dots, n$; a common variance, $\mathrm{Var}(\epsilon_i) = \sigma^2$, $i = 1, \dots, n$; and are uncorrelated, $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$, $i \neq j$. In matrix form, these assumptions can be written as

$$E\epsilon = 0 \text{ and } \mathrm{Cov}(\epsilon) = \sigma^2 I.$$

Then

$$EY = X\beta = \xi \text{ and } \mathrm{Cov}(Y) = \sigma^2 I.$$

Any estimator of the form $a'Y = a_1Y_1 + \cdots + a_nY_n$, with $a \in \mathbb{R}^n$ a constant vector, is called a *linear estimate*. Using (1.15),

$$\begin{aligned}\mathrm{Var}(a'Y) &= \mathrm{Cov}(a'Y) = a'\mathrm{Cov}(Y)a \\ &= a'(\sigma^2 I)a = \sigma^2 a'a = \sigma^2\|a\|^2. \end{aligned} \tag{14.15}$$

Because $EY = \xi$, the estimator $a'\hat{\xi}$ is unbiased for $a'\xi$. Since $\hat{\xi} = PY$, $a'\hat{\xi} = a'PY = (Pa)'Y$, and so by (14.15)

$$\mathrm{Var}(a'\hat{\xi}) = \sigma^2\|Pa\|^2. \tag{14.16}$$

Also, by (1.15) and since P is symmetric with $P^2 = P$,

$$\mathrm{Cov}(\hat{\xi}) = \mathrm{Cov}(PY) = P\mathrm{Cov}(Y)P = P(\sigma^2 I)P = \sigma^2 P.$$

When X has full rank, we can compute the covariance of the least squares estimator $\hat{\beta}$ using (1.15) as

$$\begin{aligned}\mathrm{Cov}(\hat{\beta}) &= \mathrm{Cov}\big((X'X)^{-1}X'Y\big) \\ &= (X'X)^{-1}X'\mathrm{Cov}(Y)X(X'X)^{-1} = \sigma^2(X'X)^{-1}. \end{aligned} \tag{14.17}$$

Theorem 14.4 (Gauss–Markov). *Suppose*

$$EY = X\beta \text{ and } \operatorname{Cov}(Y) = \sigma^2 I.$$

Then the (least squares) estimator $a'\hat{\xi}$ of $a'\xi$ is unbiased and has uniformly minimum variance among all linear unbiased estimators.

Proof. Let $\delta = b'Y$ be a competing unbiased estimator. By (14.15) and (14.16), the variances of δ and $a'\hat{\xi}$ are

$$\operatorname{Var}(\delta) = \sigma^2 \|b\|^2 \text{ and } \operatorname{Var}(a'\hat{\xi}) = \sigma^2 \|Pa\|^2.$$

If ϵ happens to come from a normal distribution, since both of these estimators are unbiased and $a'\hat{\xi}$ is UMVU, $\operatorname{Var}(a'\hat{\xi}) \le \operatorname{Var}(\delta)$, or

$$\sigma^2 \|Pa\|^2 \le \sigma^2 \|b\|^2.$$

But formulas for the variances of the estimators do not depend on normality, and thus $\operatorname{Var}(a'\hat{\xi}) \le \operatorname{Var}(\delta)$ in general. ☐

Although $a'\hat{\xi}$ is the "best" linear estimate, in some examples nonlinear estimates can be more precise.

Example 14.5. Suppose

$$Y_i = \beta + \epsilon_i, \qquad i = 1, \dots, n,$$

where $\epsilon_1, \dots, \epsilon_n$ are i.i.d. with common density

$$f(x) = \frac{e^{-\sqrt{2}|x|/\sigma}}{\sigma\sqrt{2}}, \qquad x \in \mathbb{R}.$$

By the symmetry, $E\epsilon_i = 0$, $i = 1, \dots, n$, and

$$\begin{aligned} \operatorname{Var}(\epsilon_i) = E\epsilon_i^2 &= 2\int_0^\infty \frac{x^2 e^{-\sqrt{2}x/\sigma}}{\sigma\sqrt{2}}\, dx \\ &= \frac{\sigma^2}{2}\int_0^\infty u^2 e^{-u}\, du = \frac{\sigma^2}{2}\Gamma(3) = \sigma^2, \qquad i = 1, \dots, n. \end{aligned}$$

So $\operatorname{Cov}(Y) = \operatorname{Cov}(\epsilon) = \sigma^2 I$, and if we take $X = (1, \dots, 1)'$, then $EY = X\beta$. This shows that the conditions of the Gauss–Markov theorem are satisfied. If $a = n^{-1}X$, then $a'\xi = n^{-1}X'X\beta = \beta$. By the Gauss–Markov theorem, the best linear estimator of β is

$$\hat{\beta} = \frac{1}{n}X'\hat{\xi} = \frac{1}{n}X'X(X'X)^{-1}X'Y = \frac{1}{n}X'Y = \overline{Y},$$

the sample average. This estimator has variance σ^2/n. A competing estimator might be the sample median,

$$\tilde{Y} = \operatorname{med}\{Y_1, \dots, Y_n\} = \beta + \operatorname{med}\{\epsilon_1, \dots, \epsilon_n\}.$$

By (8.5), $\sqrt{n}(\tilde{Y} - \beta) \Rightarrow N(0, \sigma^2/2)$. This result suggests that

$$\operatorname{Var}\bigl(\sqrt{n}(\tilde{Y} - \beta)\bigr) \to \sigma^2/2.$$

This can be established formally using Theorem 8.16 and showing that the variables $n(\tilde{Y} - \beta)^2$ are uniformly integrable. Since $\operatorname{Var}\bigl(\sqrt{n}(\overline{Y} - \beta)\bigr) = \sigma^2$, for large n the variance of $\tilde{Y}$ is roughly half the variance of $\overline{Y}$.

14.4 Estimating σ^2

From the discussion in Section 14.1, $Z_{r+1}, \dots, Z_n$ are i.i.d. from $N(0, \sigma^2)$. Thus $EZ_i^2 = \sigma^2$, $i = r + 1. \dots, n$, and the average of these variables,

$$S^2 = \frac{1}{n - r} \sum_{i=r+1}^{n} Z_i^2, \tag{14.18}$$

is an unbiased estimator of σ^2. But S^2 is a function of the complete sufficient statistic $(Z_1, \dots, Z_r, \sum_{i=1}^n Z_i^2)$ in (14.7), and so S^2 is the UMVU estimator of σ^2. The estimator S^2 can be computed from the length of the residual vector e in (14.10). To see this, first write

$$\|e\|^2 = e'e = \left(\sum_{i=r+1}^{n} Z_i v_i'\right)\left(\sum_{j=r+1}^{n} Z_j v_j\right) = \sum_{i=r+1}^{n} \sum_{j=r+1}^{n} Z_i Z_j v_i' v_j.$$

Because $v_1, \dots, v_n$ is an orthonormal basis, $v_i' v_j$ equals zero when $i \neq j$ and equals one when $i = j$. So the double summation in this equation simplifies giving

$$\|e\|^2 = \sum_{i=r+1}^{n} Z_i^2, \tag{14.19}$$

and so

$$S^2 = \frac{\|e\|^2}{n - r} = \frac{\|Y - \hat{\xi}\|^2}{n - r}. \tag{14.20}$$

Because $\hat{\xi}$ in (14.8) is a function of $Z_1, \dots, Z_r$, and e in (14.10) is a function of $Z_{r+1}, \dots, Z_n$, S^2 and $\hat{\xi}$ are independent. Also, using (14.19), (14.20), and the definition of the chi-square distribution,

$$\frac{(n - r)S^2}{\sigma^2} = \sum_{i=r+1}^{n} (Z_i/\sigma)^2 \sim \chi^2_{n-r}, \tag{14.21}$$

since $Z_i/\sigma \sim N(0, 1)$.

The distribution theory just presented can be used to set confidence intervals for linear estimators. If a is a constant vector in $\mathbb{R}^n$, then from (14.16) the standard deviation of least squares estimator $a'\hat{\xi}$ of $a'\xi$ is $\sigma\|Pa\|$. This standard deviation is naturally estimated as

$$\hat{\sigma}_{a'\hat{\xi}} \stackrel{\text{def}}{=} S\|Pa\|.$$

Theorem 14.6. *In the general linear model with $Y \sim N(\xi, \sigma^2 I)$, $\xi \in \omega$, and $\sigma^2 > 0$,*

$$\bigl(a'\hat{\xi} - \hat{\sigma}_{a'\hat{\xi}} t_{\alpha/2,n-r}, a'\hat{\xi} + \hat{\sigma}_{a'\hat{\xi}} t_{\alpha/2,n-r}\bigr)$$

is a $1-\alpha$ confidence interval for $a'\xi$.

Proof. Because $a'\hat{\xi} \sim N\bigl(a'\xi, \sigma^2\|Pa\|^2\bigr)$,

$$\frac{a'\hat{\xi} - a'\xi}{\sigma\|Pa\|} \sim N(0,1).$$

This variable is independent of $(n-r)S^2/\sigma^2$ because S^2 and $\hat{\xi}$ are independent. Using (9.2), the definition of the t-distribution,

$$\frac{\dfrac{a'\hat{\xi} - a'\xi}{\sigma\|Pa\|}}{\sqrt{\dfrac{1}{n-r}\dfrac{(n-r)S^2}{\sigma^2}}} = \frac{a'\hat{\xi} - a'\xi}{S\|Pa\|} \sim t_{n-r}.$$

The coverage probability of the stated interval is

$$\begin{aligned} P\bigl(a'\hat{\xi} - S\|Pa\| t_{\alpha/2,n-r} < a'\xi < a'\hat{\xi} + S\|Pa\| t_{\alpha/2,n-r}\bigr) \\ = P\left(-t_{\alpha/2,n-r} < \frac{a'\hat{\xi} - a'\xi}{S\|Pa\|} < t_{\alpha/2,n-r}\right) \\ = 1-\alpha. \end{aligned}$$

□

When X has full rank, β_i is a linear function of ξ, estimated by $\hat{\beta}_i$ with variance $\sigma\bigl[(X'X)^{-1}\bigr]_{ii}$. So the estimated standard deviation of $\hat{\beta}_i$ is

$$\hat{\sigma}_{\hat{\beta}_i} = S\sqrt{\bigl[(X'X)^{-1}\bigr]_{ii}},$$

and

$$\bigl(\hat{\beta}_i - \hat{\sigma}_{\hat{\beta}_i} t_{\alpha/2,n-p}, \hat{\beta}_i + \hat{\sigma}_{\hat{\beta}_i} t_{\alpha/2,n-p}\bigr) \tag{14.22}$$

is a $1-\alpha$ confidence interval for β_i.

14.5 Simple Linear Regression

To illustrate the ideas developed, we consider simple linear regression in which a response variable Y is a linear function of an independent variable x plus random error. Specifically,

$$Y_i = \beta_1 + \beta_2(x_i - \overline{x}) + \epsilon_i, \qquad i = 1, \dots, n.$$

The independent variables $x_1, \dots, x_n$ with average $\overline{x}$ are taken to be known constants, β_1 and β_2 are unknown parameters, and $\epsilon_1, \dots, \epsilon_n$ are i.i.d. from $N(0, \sigma^2)$. This gives a general linear model with design matrix

$$X = \begin{pmatrix} 1 & x_1 - \overline{x} \\ \vdots & \vdots \\ 1 & x_n - \overline{x} \end{pmatrix}.$$

In parameterizing the mean of Y (called the *regression function*) as $\beta_1 + \beta_2(x - \overline{x})$, β_1 would be interpreted not as an intercept, but as the value of the regression when $x = \overline{x}$. Note that $\sum_{i=1}^n (x_i - \overline{x}) = \sum_{i=1}^n x_i - n\overline{x} = 0$, which means that the two columns of X are orthogonal. This will simplify many later results. For instance, X will have rank 2 unless all entries in the second column are zero, which can only occur if $x_1 = \cdots = x_n$. Also, since the entries in $X'X$ are inner products of the columns of X, this matrix and $(X'X)^{-1}$ are both diagonal:

$$X'X = \begin{pmatrix} n & 0 \\ 0 & \sum_{i=1}^n (x_i - \overline{x})^2 \end{pmatrix}$$

and

$$(X'X)^{-1} = \begin{pmatrix} 1/n & 0 \\ 0 & 1/\sum_{i=1}^n (x_i - \overline{x})^2 \end{pmatrix}.$$

Since

$$X'Y = \begin{pmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n Y_i(x_i - \overline{x}) \end{pmatrix},$$

$$\hat\beta = (X'X)^{-1}X'Y = \begin{pmatrix} \frac{1}{n}\sum_{i=1}^n Y_i \\ \sum_{i=1}^n Y_i(x_i - \overline{x}) / \sum_{i=1}^n (x_i - \overline{x})^2 \end{pmatrix}.$$

Also,

$$\operatorname{Cov}(\hat\beta) = \sigma^2(X'X)^{-1} = \begin{pmatrix} \sigma^2/n & 0 \\ 0 & \sigma^2/\sum_{i=1}^n (x_i - \overline{x})^2 \end{pmatrix}. \tag{14.23}$$

To estimate σ^2, since $\hat\xi_i = \hat\beta_1 + \hat\beta_2(x_i - \overline{x})$,

$$e_i = Y_i - \hat\beta_1 - \hat\beta_2(x_i - \overline{x}),$$

and then

$$S^2 = \frac{1}{n-2} \sum_{i=1}^{n} e_i^2.$$

This equation can be rewritten in various ways. For instance,

$$S^2 = \frac{1}{n-2} \sum_{i=1}^{n} (Y_i - \overline{Y})^2 (1 - \hat{\rho}^2),$$

where $\hat{\rho}$ is the *sample correlation* defined as

$$\hat{\rho} = \frac{\sum_{i=1}^{n} (Y_i - \overline{Y})(x_i - \overline{x})}{\left[\sum_{i=1}^{n} (Y_i - \overline{Y})^2 \sum_{i=1}^{n} (x_i - \overline{x})^2\right]^{1/2}}.$$

This equation shows that $\hat{\rho}^2$ may be viewed as the proportion of the variation of Y that is "explained" by the linear relation between Y and x.

Using (14.22),

$$\left(\hat{\beta}_1 - \frac{St_{\alpha/2,n-2}}{\sqrt{n}}, \hat{\beta}_1 + \frac{St_{\alpha/2,n-2}}{\sqrt{n}}\right)$$

is a $1 - \alpha$ confidence interval for β_1, and

$$\left(\hat{\beta}_2 - \frac{St_{\alpha/2,n-2}}{\sqrt{\sum_{i=1}^{n} (x_i - \overline{x})^2}}, \hat{\beta}_2 + \frac{St_{\alpha/2,n-2}}{\sqrt{\sum_{i=1}^{n} (x_i - \overline{x})^2}}\right)$$

is a $1 - \alpha$ confidence interval for β_2.

14.6 Noncentral *F* and Chi-Square Distributions

Distribution theory for testing in the general linear model relies on noncentral F and chi-square distributions.

Definition 14.7. *If $Z_1, \dots, Z_p$ are independent and $\delta \geq 0$ with*

$$Z_1 \sim N(\delta, 1) \;\; \textit{and} \;\; Z_j \sim N(0, 1), \quad j = 2, \dots, p,$$

then $W = \sum_{i=1}^{p} Z_i^2$ has the noncentral chi-square distribution *with noncentrality parameter δ^2 and p degrees of freedom, denoted*

$$W \sim \chi_p^2(\delta^2).$$

Lemma 14.8. *If $Z \sim N_p(\mu, I)$, then $Z'Z \sim \chi_p^2(\|\mu\|^2)$.*

Proof. Let O be an orthogonal matrix where the first row is $\mu'/\|\mu\|$, so that

$$O\mu = \tilde{\mu} = \begin{pmatrix} \|\mu\| \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Then

$$\tilde{Z} = OZ \sim N_p(\tilde{\mu}, I_p).$$

From the definition, $\tilde{Z}'\tilde{Z} = \sum_{i=1}^p \tilde{Z}^2 \sim \chi_p^2(\|\mu\|^2)$, and the lemma follows because

$$\tilde{Z}'\tilde{Z} = Z'O'OZ = Z'Z. \qquad \square$$

The next lemma shows that certain quadratic forms for multivariate normal vectors have noncentral chi-square distributions.

Lemma 14.9. *If Σ is a $p \times p$ positive definite matrix and if $Z \sim N_p(\mu, \Sigma)$, then*

$$Z'\Sigma^{-1}Z \sim \chi_p^2(\mu'\Sigma^{-1}\mu).$$

Proof. Let $A = \Sigma^{-1/2}$, the symmetric square root of Σ^{-1}, defined in (9.11). Then $AZ \sim N_p(A\mu, I_p)$, and so

$$Z'\Sigma^{-1}Z = (AZ)'(AZ) \sim \chi_p^2(\|A\mu\|^2).$$

The lemma follows because $\|A\mu\|^2 = (A\mu)'(A\mu) = \mu'AA\mu = \mu'\Sigma^{-1}\mu$. $\square$

Definition 14.10. *If V and W are independent variables with $V \sim \chi_k^2(\delta^2)$ and $W \sim \chi_m^2$, then*

$$\frac{V/k}{W/m} \sim F_{k,m}(\delta^2),$$

the noncentral F-distribution *with degrees of freedom k and m and noncentrality parameter δ^2. When $\delta^2 = 0$ this distribution is simply called the F distribution, $F_{k,m}$.*

14.7 Testing in the General Linear Model

In the general linear model, $Y \sim N(\xi, \sigma^2 I)$ with the mean ξ in a linear subspace ω with dimension r. In this section we consider testing $H_0 : \xi \in \omega_0$ versus $H_1 : \xi \in \omega - \omega_0$ with ω_0 a q-dimensional linear subspace of ω, $0 \le q < r$. Null hypotheses of this form arise when β satisfies linear constraints. For instance we might have $H_0 : \beta_1 = \beta_2$ or $H_0 : \beta_1 = 0$. (Similar ideas can be used to test $\beta_1 = c$ or other affine constraints; see Problem 14.13.)

Let $\hat{\xi}$ and $\hat{\xi}_0$ denote least squares estimates for ξ under the full model and under H_0. Specifically, $\hat{\xi} = PY$ and $\hat{\xi}_0 = P_0 Y$, where P and P_0 are the projection matrices onto ω and ω_0. The test statistic of interest is based on $\|Y - \hat{\xi}\|$, the distance between Y and ω, and $\|Y - \hat{\xi}_0\|$, the distance between Y and ω_0. Because $\omega_0 \subset \omega$, the former distance must be smaller, but if the distances are comparable, then at least qualitatively H_0 may seem adequate. The test statistic is

$$T = \frac{n-r}{r-q} \frac{\|Y - \hat{\xi}_0\|^2 - \|Y - \hat{\xi}\|^2}{\|Y - \hat{\xi}\|^2},$$

and H_0 will be rejected if T exceeds a suitable constant. Noting that $Y - \hat{\xi} \in \omega^\perp$ and $\hat{\xi} - \hat{\xi}_0 \in \omega$, the vectors $Y - \hat{\xi}$ and $\hat{\xi} - \hat{\xi}_0$ are orthogonal, and the squared length of their sum, by the Pythagorean theorem, is

$$\|Y - \hat{\xi}_0\|^2 = \|Y - \hat{\xi}\|^2 + \|\hat{\xi} - \hat{\xi}_0\|^2.$$

Using this, the formula for T can be rewritten as

$$T = \frac{n-r}{r-q} \frac{\|\hat{\xi} - \hat{\xi}_0\|^2}{\|Y - \hat{\xi}\|^2} = \frac{\|\hat{\xi} - \hat{\xi}_0\|^2}{(r-q)S^2}. \tag{14.24}$$

This test statistic is equivalent to the generalized likelihood ratio test statistic that will be introduced and studied in Chapter 17. When $r - q = 1$ the test is uniformly most powerful unbiased, and when $r - q > 1$ the test is most powerful among tests satisfying symmetry restrictions.

For level and power calculations we need the distribution of T given in the next theorem.

Theorem 14.11. *Under the general linear model,*

$$T \sim F_{r-q,n-r}(\delta^2),$$

where

$$\delta^2 = \frac{\|\xi - P_0\xi\|^2}{\sigma^2}. \tag{14.25}$$

Proof. Write

$$Y = \sum_{i=1}^{n} Z_i v_i,$$

where $v_1, \dots, v_n$ is an orthonormal basis chosen so that $v_1, \dots, v_q$ span ω_0 and $v_1, \dots, v_r$ span ω. Then, as in (14.8),

$$\hat{\xi}_0 = \sum_{i=1}^{q} Z_i v_i \text{ and } \hat{\xi} = \sum_{i=1}^{r} Z_i v_i.$$

Also, as in (14.5) and (14.6), $Z \sim N(\eta, \sigma^2 I)$ with $\eta_{r+1} = \cdots = \eta_n = 0$. Since $v_i'v_j$ is zero when $i \neq j$ and one when $i = j$,

$$\begin{aligned}\|Y - \hat{\xi}\|^2 = \left\|\sum_{i=r+1}^n Z_i v_i\right\|^2 &= \left(\sum_{i=r+1}^n Z_i v_i'\right)\left(\sum_{j=r+1}^n Z_j v_j\right)\\ &= \sum_{i=r+1}^n \sum_{j=r+1}^n Z_i Z_j v_i' v_j = \sum_{i=r+1}^n Z_i^2.\end{aligned}$$

Similarly

$$\|Y - \hat{\xi}_0\|^2 = \sum_{i=q+1}^n Z_i^2,$$

and so

$$T = \frac{\dfrac{1}{r-q}\sum_{i=q+1}^r (Z_i/\sigma)^2}{\dfrac{1}{n-r}\sum_{i=r+1}^n (Z_i/\sigma)^2}.$$

The variables Z_i are independent, and so the numerator and denominator in this formula for T are independent. Because $Z_i/\sigma \sim N(\eta_i/\sigma, 1)$, by Lemma 14.8,

$$\sum_{i=q+1}^r \left(\frac{Z_i}{\sigma}\right)^2 \sim \chi^2_{r-q}(\delta^2),$$

where

$$\delta^2 = \sum_{i=q+1}^r \frac{\eta_i^2}{\sigma^2}. \tag{14.26}$$

Also, since $\eta_i = 0$ for $i = r+1, \dots, n$, $Z_i/\sigma \sim N(0, 1)$, $i = r+1, \dots, n$, and so $\sum_{i=r+1}^n (Z_i/\sigma)^2 \sim \chi^2_{n-r}$. So by Definition 14.10 for the noncentral F-distribution, $T \sim F_{r-q,n-r}(\delta^2)$ with δ^2 given in (14.26). To finish we must show that (14.25) and (14.26) agree, or that

$$\sum_{i=q+1}^r \eta_i^2 = \|\xi - P_0\xi\|^2.$$

Since

$$\xi = E\hat{\xi} = \sum_{i=1}^r \eta_i v_i$$

and

$$P_0\xi = EP_0Y = E\hat{\xi}_0 = \sum_{i=1}^q \eta_i v_i,$$

$$\xi - P_0\xi = \sum_{i=q+1}^{r} \eta_i v_i.$$

Then, by the Pythagorean theorem,

$$\|\xi - P_0\xi\|^2 = \sum_{i=q+1}^{r} \eta_i^2,$$

as desired. ☐

Example 14.12. Consider a model in which

$$Y_i = \begin{cases} x_i\beta_1 + \epsilon_i, & i = 1, \dots, n_1; \\ x_i\beta_2 + \epsilon_i, & i = n_1 + 1, \dots, n_1 + n_2 = n, \end{cases}$$

with $\epsilon_1, \dots, \epsilon_n$ i.i.d. from $N(0, \sigma^2)$ and independent variables $x_1, \dots, x_n$ taken as known constants. This model might be appropriate if you have bivariate data from two populations, each satisfying simple linear regression through the origin. In such a situation, the most interesting hypothesis to consider may be that the two populations are the same, and so let us test $H_0 : \beta_1 = \beta_2$ versus $H_1 : \beta_1 \neq \beta_2$. The design matrix under the full model is

$$X = \begin{pmatrix} x_1 & 0 \\ \vdots & \vdots \\ x_{n_1} & 0 \\ 0 & x_{n_1+1} \\ \vdots & \vdots \\ 0 & x_n \end{pmatrix},$$

and straightforward algebra gives

$$X'X = \begin{pmatrix} \sum_{i=1}^{n_1} x_i^2 & 0 \\ 0 & \sum_{i=n_1+1}^{n} x_i^2 \end{pmatrix} \text{ and } X'Y = \begin{pmatrix} \sum_{i=1}^{n_1} x_iY_i \\ \sum_{i=n_1+1}^{n} x_iY_i \end{pmatrix}.$$

So

$$\hat\beta = (X'X)^{-1}X'Y = \begin{pmatrix} \sum_{i=1}^{n_1} x_iY_i / \sum_{i=1}^{n_1} x_i^2 \\ \sum_{i=n_1+1}^{n} x_iY_i / \sum_{i=n_1+1}^{n} x_i^2 \end{pmatrix}$$

and

$$\|Y - \hat\xi\|^2 = \sum_{i=1}^{n_1}(Y_i - x_i\hat\beta_1)^2 + \sum_{i=n_1+1}^{n}(Y_i - x_i\hat\beta_2)^2.$$

Here X is of full rank (unless $x_1 = \cdots = x_{n_1} = 0$ or $x_{n_1} = \cdots = x_n = 0$), and so $r = p = 2$ and

$$S^2 = \frac{\|Y - \hat{\xi}\|^2}{n-2}.$$

Under H_0 we also have a general linear model. If β_0 denotes the common value of β_1 and β_2, then $Y_i = x_i\beta_0 + \epsilon_i$, $i = 1, \dots, n$, and the design matrix is

$$X_0 = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Then $X_0'X_0 = \sum_{i=1}^n x_i^2$ and $X_0'Y = \sum_{i=1}^n x_iY_i$, and so

$$\hat{\beta}_0 = (X_0'X_0)^{-1}X_0'Y = \frac{\sum_{i=1}^n x_iY_i}{\sum_{i=1}^n x_i^2}.$$

Thus

$$\begin{aligned}\|\hat{\xi} - \hat{\xi}_0\|^2 &= \sum_{i=1}^{n_1}(x_i\hat{\beta}_1 - x_i\hat{\beta}_0)^2 + \sum_{i=n_1+1}^{n}(x_i\hat{\beta}_2 - x_i\hat{\beta}_0)^2 \\ &= (\hat{\beta}_1 - \hat{\beta}_0)^2\sum_{i=1}^{n_1}x_i^2 + (\hat{\beta}_2 - \hat{\beta}_0)^2\sum_{i=n_1+1}^{n}x_i^2.\end{aligned}$$

Noting that $\hat{\beta}_0$ is a weighted average of $\hat{\beta}_1$ and $\hat{\beta}_2$,

$$\hat{\beta}_0 = \frac{\sum_{i=1}^{n_1}x_i^2}{\sum_{i=1}^n x_i^2}\hat{\beta}_1 + \frac{\sum_{i=n_1+1}^{n}x_i^2}{\sum_{i=1}^n x_i^2}\hat{\beta}_2,$$

this expression simplifies to

$$\|\hat{\xi} - \hat{\xi}_0\|^2 = \frac{\sum_{i=1}^{n_1}x_i^2\sum_{i=n_1+1}^{n}x_i^2}{\sum_{i=1}^n x_i^2}(\hat{\beta}_1 - \hat{\beta}_2)^2. \tag{14.27}$$

So the test statistic T is given by

$$T = \frac{\sum_{i=1}^{n_1}x_i^2\sum_{i=n_1+1}^{n}x_i^2}{S^2\sum_{i=1}^n x_i^2}(\hat{\beta}_1 - \hat{\beta}_2)^2.$$

Under $H_0, T \sim F_{1,n-2}$, and the level-α test will reject H_0 if T exceeds $F_{\alpha,1,n-2}$, the upper αth quantile of this F-distribution.

For power calculations we need the noncentrality parameter δ^2 in (14.25). Given the calculations we have done, the easiest way to find δ^2 is to note that if our data were observed without error, i.e., if ϵ were zero, then $\hat{\xi}$ would be ξ, $\hat{\beta}_1$ and $\hat{\beta}_2$ would be β_1 and β_2, and $\hat{\xi}_0$ would be $P_0\xi$. Using this observation and (14.27),

$$\delta^2 = \frac{\sum_{i=1}^{n_1}x_i^2\sum_{i=n_1+1}^{n}x_i^2}{\sigma^2\sum_{i=1}^n x_i^2}(\beta_1 - \beta_2)^2.$$

The power is then the chance that a variable from $F_{1,n-2}(\delta^2)$ exceeds $F_{\alpha,1,n-2}$.

14.8 Simultaneous Confidence Intervals

A researcher studying a complex data set may construct confidence intervals for many parameters. Even with a high coverage probability for each interval there may be a substantial chance some of them will fail, raising a concern that the ones that fail may be reported as meaningful when all that is really happening is natural chance variation. Simultaneous confidence intervals have been suggested to protect against this possibility. A few basic ideas are developed here, first in the context of one-way ANOVA models, introduced before in Example 14.2.

The model under consideration has

$$Y_{kl} = \beta_k + \epsilon_{kl}, \qquad 1 \le l \le c, \qquad 1 \le k \le p.$$

This can be viewed as a model for independent random samples from p normal populations all with a common variance. The design here has the same number of observations c from each population. Listing the variables Y_{kl} in a single vector, as in Example 14.2, this is a general linear model. The least squares estimator of β should minimize

$$\sum_{l=1}^{c}\sum_{k=1}^{p}(Y_{kl} - \beta_k)^2.$$

The partial derivative of this criterion with respect to β_m is

$$-2\sum_{l=1}^{c}(Y_{ml} - \beta_m)$$

which vanishes when $\beta_m = \hat\beta_m$ given by

$$\hat\beta_m = \overline{Y}_{m\cdot} \stackrel{\text{def}}{=} \frac{1}{c}\sum_{l=1}^{c} Y_{ml}, \qquad m = 1, \dots, p.$$

These are the least squares estimators. Here $r = p$ and $n = pc$, so

$$S^2 = \frac{\|Y - \hat\xi\|^2}{pc - p} = \frac{1}{p(c-1)}\sum_{l=1}^{c}\sum_{k=1}^{p}(Y_{kl} - \hat\beta_k)^2.$$

The least squares estimators are averages of different collections of the Y_{kl}. Thus $\hat\beta_1, \dots, \hat\beta_p$ are independent with

$$\hat\beta_k \sim N(\beta_k, \sigma^2/c), \qquad k = 1, \dots, p.$$

Also

$$\frac{p(c-1)S^2}{\sigma^2} \sim \chi^2_{p(c-1)},$$

and S^2 is independent of $\hat{\beta}$.

To start, let us try to find intervals $I_1, \dots, I_p$ that simultaneously cover parameters $\beta_1, \dots, \beta_p$ with specified probability $1-\alpha$. Specifically, we want

$$P(\beta_k \in I_k, k = 1, \dots, p) = 1 - \alpha.$$

The confidence intervals in Theorem 14.6 or (14.22) are

$$\left(\hat{\beta}_k - \frac{S}{\sqrt{c}} t_{\alpha/2, p(c-1)}, \hat{\beta}_k + \frac{S}{\sqrt{c}} t_{\alpha/2, p(c-1)}\right),$$

and intuition suggests that we may be able to achieve our objective taking

$$I_k = \left(\hat{\beta}_k - \frac{S}{\sqrt{c}} q, \hat{\beta}_k + \frac{S}{\sqrt{c}} q\right), \qquad k = 1, \dots, p,$$

if q is chosen suitably. Now

$$\begin{aligned} P(\beta_k \in I_k, k = 1, \dots, p) &= P\left(|\hat{\beta}_k - \beta_k| < \frac{S}{\sqrt{c}} q, k = 1, \dots, p\right) \\ &= P\left(\max_{1 \le k \le p} \frac{|\hat{\beta}_k - \beta_k|}{S/\sqrt{c}} < q\right) \\ &= P\left(\max_{1 \le k \le p} \frac{|Z_k|}{\sqrt{W}} < q\right), \end{aligned}$$

where

$$Z_k = \frac{\hat{\beta}_k - \beta_k}{\sigma/\sqrt{c}} \sim N(0,1), \qquad k = 1, \dots, p$$

and $W = S^2/\sigma^2$. Because $Z_1, \dots, Z_p$ and W are independent and $mW \sim \chi^2_m$ with $m = p(c-1)$, the simultaneous coverage probability here does not depend on β or σ.

Definition 14.13. *If $Z_1, \dots, Z_p$ and W are independent variables with $Z_k \sim N(0,1)$, $k = 1, \dots, p$, and $mW \sim \chi^2_m$, then*

$$\frac{\max_{1 \le k \le p} |Z_k|}{\sqrt{W}}$$

has the studentized maximum modulus distribution *with parameters p and m.*

If q is the upper αth quantile of this studentized maximum modulus distribution, then the intervals $I_1, \dots, I_p$ have simultaneous coverage probability $1 - \alpha$.

In practice, researchers will often be more interested in comparing populations than estimating individual means, and so confidence intervals for differences $\beta_j - \beta_i$ may be of interest. So let us now seek intervals I_{ij} such that

$$P(\beta_j - \beta_i \in I_{ij}, \forall i \neq j) = 1 - \alpha.$$

Now we may naturally suspect we can achieve this objective with intervals of the form

$$I_{ij} = \left(\hat{\beta}_j - \hat{\beta}_i - \frac{S}{\sqrt{c}}q, \hat{\beta}_j - \hat{\beta}_i + \frac{S}{\sqrt{c}}q\right)$$

with q adjusted suitably. Then

$$\begin{aligned}
P(\beta_j &- \beta_i \in I_{ij}, \forall i \neq j) \\
&= P\left(\left|(\hat{\beta}_j - \beta_j) - (\hat{\beta}_i - \beta_i)\right| < \frac{S}{\sqrt{c}}q, \forall i \neq j\right) \\
&= P\left(\frac{\left|\sqrt{c}(\hat{\beta}_j - \beta_j) - \sqrt{c}(\hat{\beta}_i - \beta_i)\right|}{S} < q, \forall i \neq j\right) \\
&= P\left(\frac{|Z_j - Z_i|}{\sqrt{W}} < q, \forall i \neq j\right) \\
&= P\left(\frac{\max_{1\le k\le p} Z_k - \min_{1\le k\le p} Z_k}{\sqrt{W}} < q\right).
\end{aligned}$$

This approach works because this probability does not depend on β or σ.

Definition 14.14. *If $Z_1, \dots, Z_p$ and W are independent variables with $Z_k \sim N(0,1)$, $k = 1, \dots, p$ and $mW \sim \chi^2_m$, then*

$$\frac{\max_{1\le k\le p} Z_k - \min_{1\le k\le p} Z_k}{\sqrt{W}}$$

has the studentized range distribution *with parameters p and m.*

If q is the upper αth quantile of the studentized range distribution, then the intervals I_{ij} will have simultaneous coverage probability $1 - \alpha$.

The derivations of simultaneous confidence intervals just presented relies heavily on the structure of the ANOVA model under consideration. More general results are possible using a method due to Scheffé. This method is based on confidence sets for a parameter $\psi \in \mathbb{R}^q$, with $q \le r$, which is a linear function of the mean ξ; that is,

$$\psi = A\xi = AX\beta$$

for some $q \times n$ matrix A. When X is full rank $\beta = (X'X)^{-1}X'\xi$, and so $A = (X'X)^{-1}X'$ will give $\psi = \beta$. Other linear functions of β are also possible. Because $P\xi = \xi$, we have $AP\xi = A\xi = \psi$, and replacing A by $A^* = AP$ will not change ψ. Then $A^*P = APP = AP = A^*$. Changing A to A^*, if necessary, we can assume without loss of generality that $A = AP$. This is convenient because then the least squares estimator of ψ is

$$\hat{\psi} = A\hat{\xi} = APY = AY.$$

Finally, we assume that the rows of AX are linearly independent. Since AX is then of full rank and $\psi = AX\beta$, this ensures that ψ can assume arbitrary values in $\mathbb{R}^q$. Also, note that then the rank of AX will be less than the rank of A, for if the rows of A satisfy a nontrivial linear constraint, $v'A = 0$, then $v'AX = 0$, and the rows of AX will satisfy the same linear constraint. If we define $B = AA'$, then B is positive definite because

$$q \le \operatorname{rank}(AX) \le \operatorname{rank}(A) \le q,$$

showing that A and AX are both of full rank, and $v'Bv = v'AA'v = \|A'v\|^2$, which is positive unless $v = 0$ as A is full rank. Using (1.15),

$$\hat{\psi} \sim N(\psi, \sigma^2 B),$$

and so by Lemma 14.9,

$$\frac{(\hat{\psi} - \psi)'B^{-1}(\hat{\psi} - \psi)}{\sigma^2} \sim \chi_q^2.$$

Because $\hat{\psi}$ is a function of $\hat{\xi}$, and $\hat{\xi}$ is independent of S^2, this quadratic form is independent of

$$\frac{(n-r)S^2}{\sigma^2} \sim \chi_{n-r}^2.$$

Then by Definition 14.10,

$$\frac{(\hat{\psi} - \psi)'B^{-1}(\hat{\psi} - \psi)/(q\sigma^2)}{S^2/\sigma^2} = \frac{(\hat{\psi} - \psi)'B^{-1}(\hat{\psi} - \psi)}{qS^2} \sim F_{q,n-r}.$$

From this

$$P\big((\hat{\psi} - \psi)'B^{-1}(\hat{\psi} - \psi) \le qS^2 F_{\alpha,q,n-r}\big) = 1 - \alpha.$$

The set of values for ψ where this event obtains is a multivariate ellipse centered at $\hat{\psi}$. This random ellipse is a $1 - \alpha$ confidence set for ψ.

To form simultaneous confidence intervals from the confidence ellipse described, note that

$$(\hat{\psi} - \psi)'B^{-1}(\hat{\psi} - \psi) = \|B^{-1/2}(\hat{\psi} - \psi)\|^2.$$

Also, for any $h \in \mathbb{R}^q$,

$$h'Bh = h'B^{1/2}B^{1/2}h = \|B^{1/2}h\|^2.$$

By the Schwarz inequality,

$$\|B^{1/2}h\|^2\|B^{-1/2}(\hat{\psi} - \psi)\|^2 \ge \big[h'B^{1/2}B^{-1/2}(\hat{\psi} - \psi)\big]^2 = \big[h'(\hat{\psi} - \psi)\big]^2.$$

So

$$\begin{aligned}
&P\Big\{\big[h'(\hat\psi-\psi)\big]^2 \le qS^2h'BhF_{\alpha,q,n-r}, \forall h\in\mathbb{R}^q\Big\}\\
&\qquad \ge P\big(\|B^{1/2}h\|^2\|B^{-1/2}(\hat\psi-\psi)\|^2 \le qS^2h'BhF_{\alpha,q,n-r}, \forall h\in\mathbb{R}^q\big)\\
&\qquad = P\big(\|B^{-1/2}(\hat\psi-\psi)\|^2 \le qS^2F_{\alpha,q,n-r}\big)\\
&\qquad = 1-\alpha.
\end{aligned}$$

But taking $h=B^{-1}(\hat\psi-\psi)$, this probability can be at most $1-\alpha$, and so we must have equality. Since

$$\mathrm{Var}\big(h'(\hat\psi-\psi)\big)=\sigma^2h'Bh,$$

naturally estimated by

$$\hat\sigma^2_{h'\hat\psi}=S^2h'Bh,$$

we can write this identity as

$$P\Big\{\big[h'(\hat\psi-\psi)\big]^2 \le \hat\sigma^2_{h'\hat\psi}qF_{\alpha,q,n-r}, \forall h\in\mathbb{R}^q\Big\}=1-\alpha.$$

So the intervals

$$\Big(h'\hat\psi-\hat\sigma_{h'\hat\psi}\sqrt{qF_{\alpha,q,n-r}},\, h'\hat\psi+\hat\sigma_{h'\hat\psi}\sqrt{qF_{\alpha,q,n-r}}\Big)$$

contain $h'\psi$ simultaneously for all $h\in\mathbb{R}^q$, with probability $1-\alpha$.

Example 14.15. In the model for simple linear regression considered in Section 14.5, the value for the regression function at a specified value x for the independent variable is $\beta_1+\beta_2(x-\overline{x})$, estimated by $\hat\beta_1+\hat\beta_2(x-\overline{x})$. Using (14.23), the variance of this estimator is

$$\sigma^2\left[\frac{1}{n}+\frac{(x-\overline{x})^2}{s_{xx}}\right],$$

where $s_{xx}=\sum_{i=1}^n(x_i-\overline{x})^2$. This variance is estimated by

$$S^2\left[\frac{1}{n}+\frac{(x-\overline{x})^2}{s_{xx}}\right].$$

Taking $\psi=\beta$, the regression at x, $\beta_1+\beta_2(x-\overline{x})$, will lie in

$$\begin{aligned}
\Bigg(&\hat\beta_1+\hat\beta_2(x-\overline{x})-S\sqrt{\frac{1}{n}+\frac{(x-\overline{x})^2}{s_{xx}}}\sqrt{2F_{\alpha,2,n-2}},\\
&\qquad\qquad \hat\beta_1+\hat\beta_2(x-\overline{x})+S\sqrt{\frac{1}{n}+\frac{(x-\overline{x})^2}{s_{xx}}}\sqrt{2F_{\alpha,2,n-2}}\Bigg),
\end{aligned}$$

simultaneously for all $x\in\mathbb{R}$, with probability at least $1-\alpha$. These confidence bands are plotted along with the regression line in Figure 14.1.

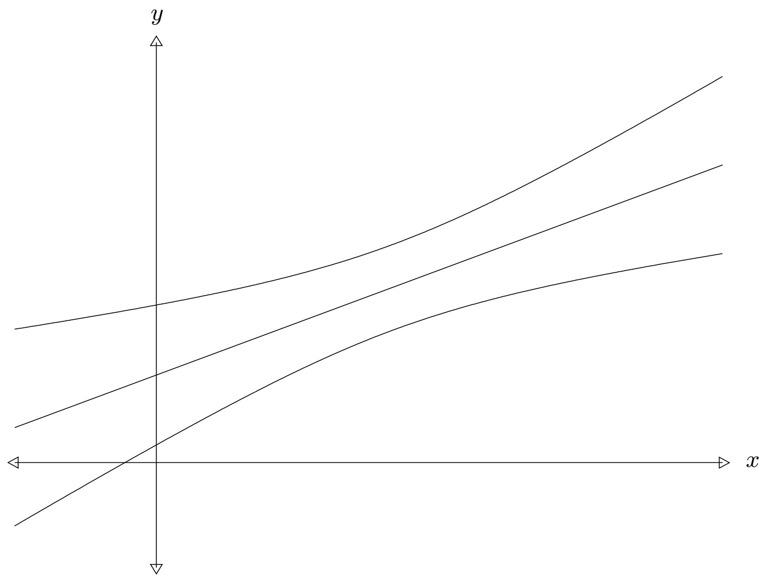

Fig. 14.1. Confidence bands.

Example 14.16. In one-way ANOVA comparisons between weighted averages of the means of certain populations and weighted averages of the means of different populations may be of interest. Scaled differences between weighted averages are called *contrasts*.

Definition 14.17. *A* contrast *in one-way ANOVA is any parameter of the form*

$$a'\beta = a_1\beta_1 + \cdots + a_p\beta_p$$

with $a_1 + \cdots + a_p = 0$.

Examples of contrasts include $\beta_3 - \beta_1$ or $\frac{1}{2}(\beta_1 + \beta_3) - \left(\frac{1}{6}\beta_2 + \frac{1}{3}\beta_4 + \frac{1}{2}\beta_5\right)$. Taking

$$\psi = \begin{pmatrix} \beta_1 - \beta_p \\ \vdots \\ \beta_{p-1} - \beta_p \end{pmatrix},$$

a contrast $a'\beta$ equals $h'\psi$ with $h' = (a_1, \dots, a_{p-1})$. The least squares estimate of this contrast is $\sum_{i=1}^p a_i\hat{\beta}_i$ which has variance $\sum_{i=1}^p a_i^2\sigma^2/c = \|a\|^2\sigma^2/c$, estimated replacing σ^2 by S^2. Thus with probability $1-\alpha$,

$$a'\beta \in \left(a'\hat{\beta} - \frac{S\|a\|}{\sqrt{c}}\sqrt{(p-1)F_{\alpha,p-1,p(c-1)}},\right.$$
$$\left. a'\hat{\beta} + \frac{S\|a\|}{\sqrt{c}}\sqrt{(p-1)F_{\alpha,p-1,p(c-1)}}\right),$$

simultaneously for all $a \in \mathbb{R}^p$ with $a_1 + \cdots + a_p = 0$. If instead we were interested in all linear combinations of β, we would need to take $\psi = \beta$. For this case, q equals p, and we have simultaneous confidence intervals

$$a'\beta \in \left(a'\hat{\beta} - \frac{S\|a\|}{\sqrt{c}}\sqrt{pF_{\alpha,p,p(c-1)}},\, a'\hat{\beta} + \frac{S\|a\|}{\sqrt{c}}\sqrt{pF_{\alpha,p,p(c-1)}} \right),$$

for all $a \in \mathbb{R}^p$. These intervals will be a bit wider than the simultaneous confidence intervals for contrasts.

14.9 Problems[1]

*1. Consider a model in which our data are

$$Y_i = \beta_1 + w_i\beta_2 + x_i\beta_3 + \epsilon_i, \qquad i = 1, \ldots, n,$$

where $w_1, \ldots, w_n$ and $x_1, \ldots, x_n$ are observed constants; β_1, β_2, and β_3 are unobserved parameters; and $\epsilon_1, \ldots, \epsilon_n$ are unobserved random variables which are i.i.d. from $N(0, \sigma^2)$. Assume that through accident or design, $w_1 + \cdots + w_n = 0$ and $x_1 + \cdots + x_n = 0$. For notation, let $S_{ww} = \sum_{i=1}^n w_i^2$, $S_{xx} = \sum_{i=1}^n x_i^2$, $S_{wx} = \sum_{i=1}^n w_i x_i$, and so on.

a) Write the model in matrix form as $Y = X\beta + \epsilon$ describing entries in the matrix X.
b) If $n > 3$, when will X be of full rank?
c) Assuming X is of full rank, give an explicit formula for $\hat{\beta}$. (It will involve terms such as S_{xx}, S_{xY}, etc.)
d) Find the covariance matrix of $\hat{\beta}$.
e) Give a formula for the UMVU estimator of σ^2.
f) Find confidence intervals for β_1 and for $\beta_3 - \beta_2$.

2. Consider a general linear model with $n = 2m$ in which $Y_i = \beta_1 + \beta_3 x_i + \epsilon_i$, $i = 1, \ldots, m$, and $Y_i = \beta_2 + \beta_3 x_i + \epsilon_i$, $i = m+1, \ldots, n$. Here $\epsilon_1, \ldots, \epsilon_n$ are i.i.d. from $N(0, \sigma^2)$; $\beta = (\beta_1, \beta_2, \beta_3)'$ and σ^2 are unknown parameters, and $x_1, \ldots, x_n$ are known constants with $x_1 + \cdots + x_m = x_{m+1} + \cdots + x_n = 0$.

a) Write the model in vector form as $Y = X\beta + \epsilon$ describing entries in the design matrix X.
b) Determine the UMVU estimator $\hat{\beta}$ of β.
c) Write β_1 as a linear function of $\xi = EY$; that is, find a vector w such that $\beta_1 = w'\xi$.
d) Find the UMVU estimator $\hat{\xi}_i$ of ξ_i.
e) Show that $\hat{\beta}_1$ from part (b) equals $w'\hat{\xi}$ from parts (c) and (d).
f) Determine the variance of $\hat{\beta}_2 - \hat{\beta}_1$.
g) Determine the UMVU estimator S^2 of σ^2.

[1] Solutions to the starred problems are given at the back of the book.

h) Derive a $1-\alpha$ confidence interval for $\beta_2 - \beta_1$.
i) If the parameter β were known, the best estimate for a future observation Y from the first population (the population for the first half of the data observed) at some specified level x for the independent variable would be $\beta_1 + \beta_3 x$. Give a 95% confidence interval for this quantity. Sketch the upper and lower limits of this interval as a function of x if $\hat{\beta}_1 = 0$ and $\hat{\beta}_3 = 1/2$.
j) Give an explicit formula for the test statistic T used to test $H_0 : \beta_1 = \beta_2$ versus $H_1 : \beta_1 \neq \beta_2$, and explain how this statistic would be used to test H_0 versus H_1 at level α.
k) Determine the distribution for T with an explicit formula for the noncentrality parameter.
l) Show that the test of $\beta_1 = \beta_2$ rejects H_0 if and only if the confidence interval for $\beta_2 - \beta_1$ does not contain zero.

3. Consider a general linear model with $n = 2m$ in which

$$Y_i = \beta_1 + \beta_2 x_i + \epsilon_i, \qquad i = 1, \dots, m,$$

and

$$Y_i = \beta_1 + \beta_3 x_i + \epsilon_i, \qquad i = m+1, \dots, n.$$

Here $\epsilon_1, \dots, \epsilon_n$ are i.i.d. from $N(0, \sigma^2)$; $\beta = (\beta_1, \beta_2, \beta_3)'$ and σ^2 are unknown parameters; and $x_1, \dots, x_n$ are known constants with $x_1 + \cdots + x_m = x_{m+1} + \cdots + x_n = 0$.
a) Write the model in vector form as $Y = X\beta + \epsilon$ describing entries in the design matrix X.
b) Determine the UMVU estimator $\hat{\beta}$ of β.
c) Write β_2 as a linear function of $\xi = EY$; that is, find a vector w such that $\beta_2 = w'\xi$.
d) Find the UMVU estimator $\hat{\xi}_i$ of ξ_i.
e) Show that $\hat{\beta}_2$ from part (b) equals $w'\hat{\xi}$ from parts (c) and (d).
f) Determine the UMVU estimator S^2 of σ^2.
g) Determine the variance of $\hat{\beta}_3 - \hat{\beta}_2$.
h) Derive a 95% confidence interval for $\beta_3 - \beta_2$.
i) If the parameter β were known, the best estimate for a future observation Y from the first population (the population for the first half of the data observed) at some specified level x for the independent variable would be $\beta_1 + \beta_2 x$. Give a 95% confidence interval for this quantity. Sketch the upper and lower limits of this interval as a function of x if $\hat{\beta}_1 = 0$ and $\hat{\beta}_2 = 1/2$.
j) Use the fact that a suitable multiple of $(\hat{\beta}_3 - \hat{\beta}_2)/S$ has a t-distribution to derive a test of $H_0 : \beta_2 = \beta_3$ versus $H_1 : \beta_2 \neq \beta_3$ with level $\alpha = 5\%$.
k) Show that this test of $H_0 : \beta_2 = \beta_3$ rejects H_0 if and only if the confidence interval for $\beta_3 - \beta_2$ does not contain zero.

l) Derive the F-test of $H_0 : \beta_2 = \beta_3$ versus $H_1 : \beta_2 \neq \beta_3$ with level $\alpha = 5\%$. Show that this test is the same as the test above based on the t-distribution.

*4. *Two-way analysis of variance without replication.* Suppose a researcher wishes to study the effects of two factors A and B on some response variable. If A and B both occur at m levels, then there are m^2 possible combinations of factors. If observations are expensive, a design in which there is a single observation for each treatment combination may be desirable. Let Y_{ij} denote the response when factor A is at level i and factor B is at level j. A common (additive) model for these data has

$$Y_{ij} = \alpha_i + \gamma_j + \epsilon_{ij},$$

where the ϵ_{ij} are i.i.d. from $N(0, \sigma^2)$. This can be considered a general linear model with $\beta = (\alpha_1, \dots, \alpha_m, \gamma_1, \dots, \gamma_m)'$.

a) Increasing every parameter α_i by an amount Δ and simultaneously decreasing every parameter γ_j by the same amount Δ leaves the mean ξ of Y unchanged. So it is clear that $r = \dim(\omega)$ is less than $p = 2m$. Determine the dimension r of ω in this model.
b) Find the least squares estimate for $\xi_{ij} = EY_{ij}$ in this model.
c) Determine S^2, the usual unbiased estimator of σ^2.
d) Show that $\alpha_i - \alpha_j$ is a linear function of ξ and determine the least squares estimate of this difference.
e) Let $\overline{Y}_{i\cdot} = (Y_{i1} + \cdots + Y_{im})/m$, the average response with factor A at level i. Use the studentized range distribution and the fact that $\overline{Y}_{1\cdot}, \dots, \overline{Y}_{m\cdot}$ are independent to derive simultaneous $1 - \alpha$ confidence intervals for all differences $\alpha_i - \alpha_j$, $1 \leq i < j \leq m$.
f) Derive a level $1 - \alpha$ test of $H_0 : \alpha_1 = \cdots = \alpha_m$ versus $H_1 : \alpha_i \neq \alpha_j$ for some $i \neq j$.
g) Give the power for this test of $\alpha_1 = \cdots = \alpha_m$. Your answer should involve the cumulative distribution function for a noncentral F distribution. Give the degrees of freedom and provide a formula for the noncentrality parameter.
h) Use the Scheffé method to derive simultaneous confidence intervals for all contrasts of the α_i, that is, all linear combinations $a_1\alpha_1 + \cdots + a_m\alpha_m$ with $a_1 + \cdots + a_m = 0$. Note that these contrasts are estimable because they can be written as linear combinations of differences $\alpha_1 - \alpha_m, \dots, \alpha_{m-1} - \alpha_m$.

5. Time series regression models often incorporate regular oscillation over time, and in some cases this structure can be incorporated into a general linear model. Let t_j, $j = 1, \dots, n$, be known time points, and assume

$$Y_j = r\sin(t_j + \theta) + \epsilon_j, \qquad j = 1, \dots, n.$$

Here the errors ϵ_i are i.i.d. from $N(0, \sigma^2)$, and $r > 0$, $\theta \in [-\pi, \pi)$, and $\sigma^2 > 0$ are unknown parameters. Assume for convenience that the time points are evenly spaced with $4k$ observations per cycle and m cycles, so that $n = 4km$ and $t_j = j\pi/(2k)$. With these assumptions,

$$\sum_{j=1}^{n} \sin^2(t_j) = \sum_{j=1}^{n} \cos^2(t_j) = n/2$$

and

$$\sum_{j=1}^{n} \sin(t_j) = \sum_{j=1}^{n} \cos(t_j) = \sum_{j=1}^{n} \sin(t_j)\cos(t_j) = 0.$$

a) Introduce new parameters $\beta_1 = r\sin\theta$ and $\beta_2 = r\cos\theta$. Show that after replacing r and θ with these parameters, we have a general linear model.
b) Find UMVU estimators $\hat\beta_1$ and $\hat\beta_2$ for β_1 and β_2.
c) Find the UMVU estimator of σ^2.
d) Derive 95% confidence intervals for β_1 and β_2.
e) Show that a suitable multiple of $\hat r^2 = \hat\beta_1^2 + \hat\beta_2^2$ has a noncentral chi-square distribution. Identify the degrees of freedom and the noncentrality parameter.
f) Derive a test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta \ne \theta_0$ with level $\alpha = 5\%$.

6. Let $\beta_1, \dots, \beta_3$ be the angles for a triangle in degrees, so $\beta_1 + \beta_2 + \beta_3 = 180$; and let $Y_1, \dots, Y_3$ be measurements of these angles. Assume that the measurement errors, $\epsilon_i = Y_i - \beta_i$, $i = 1, \dots, 3$, are i.i.d. $N(0, \sigma^2)$.
a) Find UMVU estimates $\hat\beta_1$ and $\hat\beta_2$ for β_1 and β_2.
b) Find the covariance matrix for $(\hat\beta_1, \hat\beta_2)$ and compare the variance of $\hat\beta_1$ with Y_1.
c) Find an unbiased estimator for σ^2.
d) Derive confidence intervals for β_1 and $\beta_2 - \beta_1$.

7. *Side conditions when $r < p$.* When $r < p$, different values for β will give the same mean $\xi = X\beta$, and various values for β will minimize $\|Y - X\beta\|^2$. One approach to force a unique answer is to impose side conditions on β. Because the row span and column span of a matrix are the same, the space $V \subset \mathbb{R}^p$ spanned by the *rows* of X will have dimension $r < p$, and $V^\perp$ will have dimension $p - r$.
a) Show that $\beta \in V^\perp$ if and only if $X\beta = 0$.
b) Let $\omega = \{X\beta : \beta \in \mathbb{R}^p\}$. Show that the map $\beta \rightsquigarrow X\beta$ from V to ω is one-to-one and onto.
c) Let $h_1, \dots, h_{p-r}$ be linearly independent vectors spanning $V^\perp$. Show that $\beta \in V$ if and only if $h_i \cdot \beta = 0$, $i = 1, \dots, p - r$. Equivalently, $\beta \in V$ if and only if $H\beta = 0$, where $H' = (h_1, \dots, h_{p-r})$.
d) From part (b), there should be a unique $\hat\beta$ in V with $X\hat\beta = \hat\xi$, and $\hat\beta$ will then minimize $\|Y - X\beta\|^2$ over $\beta \in V$. Using part (c), $\hat\beta$ can be characterized as the unique value minimizing $\|Y - X\beta\|^2$ over $\beta \in \mathbb{R}^p$ satisfying the side condition $H\hat\beta = 0$. Show that $\hat\beta$ minimizes

$$\left\| \begin{pmatrix} Y \\ 0 \end{pmatrix} - \begin{pmatrix} X \\ H \end{pmatrix} \beta \right\|^2$$

over $\beta \in \mathbb{R}^p$. Use this to derive an explicit equation for $\hat{\beta}$.

8. A variable Y has a log-normal distribution with parameters μ and σ^2 if $\log Y \sim N(\mu, \sigma^2)$.
   a) Find the mean and density for the log-normal distribution.
   b) If $Y_1, \dots, Y_n$ are i.i.d. from the log-normal distribution with unknown parameters μ and σ^2, find the UMVU for μ.
   c) If $Y_1, \dots, Y_n$ are i.i.d. from the log-normal distribution with parameters μ and σ^2, with σ^2 a known constant, find the UMVU for the common mean $\nu = EY_i$.
   d) In simple linear regression, $Y_1, \dots, Y_n$ are independent with $Y_i \sim N(\beta_1 + \beta_2 x_i, \sigma^2)$. In some applications this model may be inappropriate because the Y_i are positive; perhaps Y_i is the weight or volume of the ith unit. Suggest a similar model without this defect based on the log-normal distribution. Explain how you would estimate β_1 and β_2 in your model.
9. Consider the general linear model with normality:

$$Y \sim N(X\beta, \sigma^2 I), \qquad \beta \in \mathbb{R}^p, \quad \sigma^2 > 0.$$

   If the rank r of X equals p, show that $(\hat{\beta}, S^2)$ is a complete sufficient statistic.
10. Consider a regression version of the two-sample problem in which

$$Y_i = \begin{cases} \beta_1 + \beta_2 x_i + \epsilon_i, & i = 1, \dots, n_1; \\ \beta_3 + \beta_4 x_i + \epsilon_i, & i = n_1 + 1, \dots, n_1 + n_2 = n, \end{cases}$$

   with $\epsilon_1, \dots, \epsilon_n$ i.i.d. from $N(0, \sigma^2)$. Derive a $1 - \alpha$ confidence interval for $\beta_4 - \beta_2$, the difference between the two regression slopes.
11. *Inverse linear regression.* Consider the model for simple linear regression,

$$Y_i = \beta_1 + \beta_2 (x_i - \overline{x}) + \epsilon_i, \qquad i = 1, \dots, n,$$

   studied in Section 14.5.
   a) Derive a level α-test of $H_0 : \beta_2 = 0$ versus $H_1 : \beta_2 \neq 0$.
   b) Let y_0 denote a "target" value for the mean of Y. The regression line $\beta_1 + \beta_2(x - \overline{x})$ achieves this value when the independent variable x equals

$$\theta = \overline{x} + \frac{y_0 - \beta_1}{\beta_2}.$$

   Derive a level-α test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$. Hint: You may want to find a test first assuming $y_0 = 0$. After a suitable transformation, the general case should be similar.

c) Use duality to find a confidence region, first discovered by Fieller (1954), for θ. Show that this region is an interval if the test in part (a) rejects $\beta_2 = 0$.

12. Find the mean and variance of the noncentral chi-square distribution on p degrees of freedom with noncentrality parameter δ^2.

*13. Consider a general linear model $Y \sim N(\xi, \sigma^2 I_n)$, $\xi \in \omega$, $\sigma^2 > 0$ with $\dim(\omega) = r$. Define $\psi = A\xi \in \mathbb{R}^q$ where $q < r$, and assume $A = AP$ where P is the projection onto ω, so that $\hat{\psi} = A\hat{\xi} = AY$, and that A has full rank q.
   a) The F test derived in Section 14.7 allows us to test $\psi = 0$ versus $\psi \neq 0$. Modify that theory and give a level-α test of $H_0 : \psi = \psi_0$ versus $H_1 : \psi \neq \psi_0$ with ψ_0 some constant vector in $\mathbb{R}^q$. Hint: Let $Y^* = Y - \xi_0$ with $\xi_0 \in \omega$ and $A\xi_0 = \psi_0$. Then the null hypothesis will be $H_0^* : A\xi^* = 0$.
   b) In the discussion of the Sheffé method for simultaneous confidence intervals,

$$\left\{\psi : (\psi - \hat{\psi})'(AA')^{-1}(\psi - \hat{\psi}) \leq qS^2 F_{\alpha,q,n-r}\right\}$$

   was shown to be a level $1 - \alpha$ confidence ellipse for ψ. Show that this confidence region can be obtained using duality from the family of tests in part (a).

*14. *Analysis of covariance.* Suppose

$$Y_{kl} = \beta_k + \beta_0 x_{kl} + \epsilon_{kl}, \qquad 1 \leq l \leq c, \quad 1 \leq k \leq p,$$

with the ϵ_{kl} i.i.d. from $N(0, \sigma^2)$ and the x_{kl} known constants.
   a) If $\sum_{l=1}^c x_{kl} = 0$, $k = 1, \dots, c$, use the studentized maximum modulus distribution to derive simultaneous confidence intervals for $\beta_1, \dots, \beta_p$.
   b) If $\sum_{l=1}^c x_{1l} = \sum_{l=1}^c x_{2l} = \cdots = \sum_{l=1}^c x_{pl}$, use the studentized range distribution to derive simultaneous confidence intervals for all differences $\beta_i - \beta_j$, $1 \leq i < j \leq p$. Hint: The algebra will be simpler if you first reparameterize adding an appropriate multiple of β_0 to $\beta_1, \dots, \beta_p$.

*15. *Unbalanced one-way layout.* Suppose we have samples from p normal populations with common variance, but that the sample sizes from the different populations are not the same, so that

$$Y_{ij} = \beta_i + \epsilon_{ij}, \qquad 1 \leq i \leq p, \quad j = 1, \dots, n_i,$$

with the ϵ_{ij} i.i.d. from $N(0, \sigma^2)$.
   a) Derive a level-α test of $H_0 : \beta_1 = \cdots = \beta_p$ versus $H_1 : \beta_i \neq \beta_j$ for some $i \neq j$.
   b) Use the Scheffé method to derive simultaneous confidence intervals for all contrasts $a_1\beta_1 + \cdots + a_p\beta_p$ with $a_1 + \cdots + a_p = 0$.

16. *Factorial experiments.* A "2^4" experiment is a factorial experiment to study the effects of four factors, each at two levels. The experiment has

$n = 16$ as the sample size (called the number of runs), with each run one of the 16 possible combinations of the four factors. Letting "+" and "−" be shorthand for $+1$ and -1, define vectors

$$x_A' = (+,+,+,+,+,+,+,+,-,-,-,-,-,-,-,-),$$

$$x_B' = (+,+,+,+,-,-,-,-,+,+,+,+,-,-,-,-),$$
$$x_C' = (+,+,-,-,+,+,-,-,+,+,-,-,+,+,-,-),$$
$$x_D' = (+,-,+,-,+,-,+,-,+,-,+,-,+,-,+,-),$$

and let $\mathbf{1}$ denote a column of ones. A "+1" for the jth entry of one of these vectors means that factor is set to the high level on the jth run, and a "−1" means the factor is set to the low level. So, for instance, on run 5, factors A, C, and D are at the high level, and factor B is at the low level. The vector Y gives the responses for the 16 runs. In an additive model for the experiment,

$$Y = \mu\mathbf{1} + \theta_A x_A + \theta_B x_B + \theta_C x_C + \theta_D x_D + \epsilon,$$

with the unobserved error $\epsilon \sim N(0, \sigma^2 I)$. Parameters θ_A, θ_B, θ_C, and θ_D are called the main effects for the factors.

a) Find the least squares estimates for the main effects, and give the covariance matrix for these estimators.
b) Find the UMVU estimator for σ^2.
c) Derive simultaneous confidence intervals for the main effects using the studentized maximum modulus distribution.

17. Consider the 2^4 factorial experiment described in Problem 14.16. Let x_{AB} be the elementwise product of x_A with x_B,

$$x_{AB}' = (+,+,+,+,-,-,-,-,-,-,-,-,+,+,+,+),$$

and define x_{AC}, x_{AD}, x_{BC}, x_{BD}, and x_{CD} similarly. A model with two-way interactions has

$$\begin{aligned} Y = \mu\mathbf{1} &+ \theta_A x_A + \theta_B x_B + \theta_C x_C + \theta_D x_D + \theta_{AB} x_{AB} \\ &+ \theta_{AC} x_{AC} + \theta_{AD} x_{AD} + \theta_{BC} x_{BC} + \theta_{BD} x_{BD} + \theta_{CD} x_{CD} + \epsilon, \end{aligned}$$

still with $\epsilon \sim N(0, \sigma^2 I)$. The additional parameters in this model are called two-way interaction effects. For instance, θ_{BD} is the interaction effect of factors B and D.

a) Find least squares estimators for the $\binom{4}{2}$ interaction effects, and give the covariance matrix for these estimators.
b) Derive a test for the null hypothesis that all of the interaction effects are null, that is,

$$H_0 : x_{AB} = x_{AC} = x_{AD} = x_{BC} = x_{BD} = x_{CD} = 0,$$

versus the alternative that at least one of these effects is nonzero.

c) Use the Scheffé method to derive simultaneous confidence intervals for all contrasts of the two-way interaction effects.

18. *Bonferroni approach to simultaneous confidence intervals.*

a) Suppose $I_1, \dots, I_k$ are $1 - \alpha$ confidence intervals for parameters $\eta_1 = g_1(\theta), \dots, \eta_k = g_k(\theta)$, and let γ be the simultaneous coverage probability,

$$\gamma = \inf_\theta P\big[\eta_i \in I_i, \forall i = 1, \dots, k\big].$$

Use Boole's inequality (see Problem 1.7) to derive a lower bound for γ. For a fixed value α^*, what choice for α will ensure $\gamma \geq 1 - \alpha^*$?

b) Suppose the confidence intervals in part (a) are independent. In this case, what choice for α will ensure $\gamma \geq 1 - \alpha^*$?

c) Consider one-way ANOVA with $c = 6$ observations from each of $p = 4$ populations. Compare the Bonferroni approach to simultaneous estimation of the differences $\beta_i - \beta_j$, $1 \leq i < j \leq 4$, with the approach based on the studentized range distribution. Because the Bonferroni approach is conservative, the intervals should be wider. What is the ratio of the lengths when $1 - \alpha^* = 95\%$? The 95th percentile for the studentized range distribution with parameters 4 and 20 is 3.96.